

TINGSONG XIAO

xtstars1998@gmail.com

Homepage; Google Scholar

RESEARCH INTERESTS

Large Language Models, LLM Post-training, Agentic AI, Reinforcement Learning, Harness Engineering, Multimodal AI, Recommender Systems, and AI for Healthcare.

EDUCATION

Doctor of Philosophy, Ph.D. GPA: 3.86/4.00 <i>Computer Science</i> University of Florida	Aug 2022 – May 2027 Gainesville, FL, U.S.
Master of Science, M.S. GPA: 3.86/4.00 <i>Computer Science</i> University of Florida (Conferred during Ph.D. program)	Aug 2022 – Dec 2024 Gainesville, FL, U.S.
Bachelor of Engineering, B.E. GPA: 3.92/4.00 <i>Computer Science</i> University of Electronic Science and Technology of China (UESTC)	Aug 2017 – June 2021 Chengdu, China

EXPERIENCE

Applied Scientist Intern Amazon Inc.	May 2026 – Aug 2026 Bellevue, WA, U.S.
- Worked as an Applied Scientist Intern on the Alexa AI team, focusing on agentic LLMs, long-horizon reasoning, and post-training methods for improving complex task-solving capabilities in production-oriented AI systems. - Developed and evaluated reasoning and agentic workflows involving multi-step planning, tool use, and iterative decision-making, with emphasis on robustness, controllability, and task completion quality. - Built evaluation harnesses and experimental pipelines to benchmark LLM agent behaviors across complex scenarios, enabling systematic analysis of model performance, failure modes, and post-training improvements.	
Applied Researcher Intern eBay Inc.	May 2025 – Aug 2025 San Jose, CA, U.S.
- Developed a lightweight LLM for business-critical information extraction, achieving >90% F1 across 10 high-value categories while reducing inference latency by >5×, enabling scalable deployment to millions of listings. - Built an end-to-end LLM post-training framework combining supervised fine-tuning (SFT), multi-strategy knowledge distillation, and reinforcement learning from human feedback (RLHF) methods such as DPO, improving extraction accuracy, robustness, and deployment efficiency. - Designed an autonomous LLM agentic workflow to synthesize high-fidelity training data by converting raw product descriptions into structured aspect-value pairs, reducing reliance on manual labeling and improving robustness on long-tail and rare attribute distributions.	
Research on Continuous-Time Disease Progression Modeling University of Florida	May 2025 – May 2026 Gainesville, FL, U.S.
- Addressed continuous-time disease progression modeling from longitudinal EHRs to enable early risk detection and patient sub-phenotyping under irregular visit times and patient heterogeneity (e.g., type 2 diabetes and cardiovascular complications). - Proposed TD-HNODE, integrating hypergraph representation learning from clinical trajectories with a Neural ODE to learn continuous-time dynamics of comorbidities, biomarkers, and medical events. - Outperformed state-of-the-art baselines on two large-scale real-world clinical datasets in predicting health deterioration patterns. This work was published at top AI conference ICLR 2026.	
Research on Irregular Clinical Event Sequence Modeling University of Florida	Aug 2023 – May 2024 Gainesville, FL, U.S.
Addressed irregular health time-series and event sequence prediction in real-world digital health settings (wearable sensors / mobile health), where signals are long-horizon, sparse, and non-uniformly sampled.	

- Developed **XTSFormer**, an **efficient Transformer-based** Temporal Point Process model for irregular timed clinical events, which captured **multi-scale event interactions** while remaining **computationally efficient** for long event sequence.
- Achieved **5× faster** training on long sequences (up to 100k length) and improved predictive accuracy/F1 by **3.3%** on large-scale longitudinal health datasets. This work was published at top AI conference **AAAI 2025**.

SELECTED PUBLICATIONS

- **Tingsong Xiao**, et al. "Temporally Detailed Hypergraph Neural ODE for Disease Progression Modeling." *The Fourteenth International Conference on Learning Representations (ICLR 2026)*.
- **Tingsong Xiao**, et al. "XTSFormer: Cross-Temporal-Scale Transformer for Irregular-Time Event Prediction in Clinical Applications." *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025)*.
- **Tingsong Xiao**, et al. "Large Language Models for Longitudinal Electronic Health Records: A Survey." *medRxiv 2026.1.29.345087*
- **Tingsong Xiao**, et al. "Hypergraph-Based Doubly Robust Estimation for Causal Inference of Drug Combination Effects in Heart Failure Treatment." *medRxiv 2025.12.11.25342026*
- **Tingsong Xiao**, et al. "Dual-graph Learning Convolutional Networks for Interpretable Alzheimer's Disease Diagnosis." *International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, Cham, 2022. (MICCAI 2022)*.
- Zelin Xu, Yupu Zhang, **Tingsong Xiao**, et al. "CoastalBench: A Decade-Long High-Resolution Dataset to Emulate Complex Coastal Processes." *The Forty-Second International Conference on Machine Learning (ICML 2025)*.
- Yupu Zhang, Zelin Xu, **Tingsong Xiao**, et al. "DecoyDB: A Dataset for Graph Contrastive Learning in Protein-Ligand Binding Affinity Prediction." *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*.
- Wenchong He, Zhe Jiang, and **Tingsong Xiao**, et al. "A survey on uncertainty quantification methods for deep learning." *ACM Computing Surveys (2025)*.
- Zelin Xu, **Tingsong Xiao**, et al. "Spatial-Logic-Aware Weakly Supervised Learning for Flood Mapping on Earth Imagery." *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024)*.
- Wenchong He, Zhe Jiang, and **Tingsong Xiao**, et al. "A Hierarchical Spatial Transformer for Large Numbers of Point Samples in Continuous Space." *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*.
- Zelin Xu, **Tingsong Xiao**, et al. "Spatial Knowledge-Infused Hierarchical Learning: An Application in Flood Mapping on Earth Imagery." *ACM International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL 2023)*. **Best Paper Award**.
- Jing Hu, Xincheng Wang and Ziheng Liao, **Tingsong Xiao***. "L-GCN: Local Graph Neural Network for 3D Point Clouds Classification." *IEEE International Conference on Multimedia and Expo (ICME 2023)*. (* **corresponding author**).
- Lu Zeng, Hengxin Li, **Tingsong Xiao**, et al. "Graph convolutional network with sample and feature weights for Alzheimer's disease diagnosis." *Information Processing & Management (IP&M) 59.4 (2022): 102952*.
- Xiaochen Wang, **Tingsong Xiao**, et al. "MRMRP: multi-source review-based model for rating prediction." *International Conference on Database Systems for Advanced Applications (DASFAA)*. Springer, Cham, 2020.
- Xiaochen Wang, **Tingsong Xiao**, et al. "EMRM: Enhanced Multi-source Review-Based Model for Rating Prediction." *International Conference on Knowledge Science, Engineering and Management (KSEM)*. Springer, Cham, 2021.

HONORS AND AWARDS

AAAI Scholarship and Volunteer Award

Feb. 2025

To recognize outstanding contributions and active participation through academic excellence in the 39th AAAI Conference on Artificial Intelligence (AAAI-25).

SDM Doctoral Forum Award

Mar. 2024

A competitive NSF-funded venue recognizing promising data mining researchers for their intellectual merit and broader impacts.

ACM SIGSPATIAL Best Paper Award

Nov. 2023

Awarded annually to the best paper at the ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.

Gartner Group Graduate Fellowship

Apr. 2023, Apr. 2025, Apr. 2026

Recognizes outstanding graduate students at the University of Florida for academic excellence and research potential.

MICCAI 2022 Student Travel Award

Mar. 2022

To reward first author students of the highest-quality MICCAI papers.

INVITED JOURNAL AND CONFERENCE REVIEWERS

- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Neural Information Processing Systems (NeurIPS)
- Association for the Advancement of Artificial Intelligence (AAAI)
- ACM Transactions on Intelligent Systems and Technology (TIST)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Computational Social Systems (TCSS)
- IEEE Transactions on Multimedia (TMM)

OPEN-SOURCE CONTRIBUTIONS

Qwen-Agent World Model — Contributor

QwenLM | *1k+ GitHub Stars* | [Link](#)

- Identified and fixed a critical evaluation bug in multi-turn world-model inference that omitted prior action–observation history, impairing state-dependent environment simulation.
- Implemented full interaction-history reconstruction, improving the Terminal domain benchmark score from **28.6** to **61.9 (+33.3 points)**.